

API TESTING

Tool Used

Postman

Download Link

[Postman Official Download Page](#)

Objective

The objective of API testing was to validate backend request handling, authentication, error responses, JSON structure, and API validation behavior using Postman.

A demo API platform was used to test different login and validation scenarios.

API Testing Platform

Demo Website/API Used:

[ReqRes API Platform](#)

Test Cases Performed

1. Login Request Test

Purpose

To verify successful login functionality using valid credentials.

Request Details

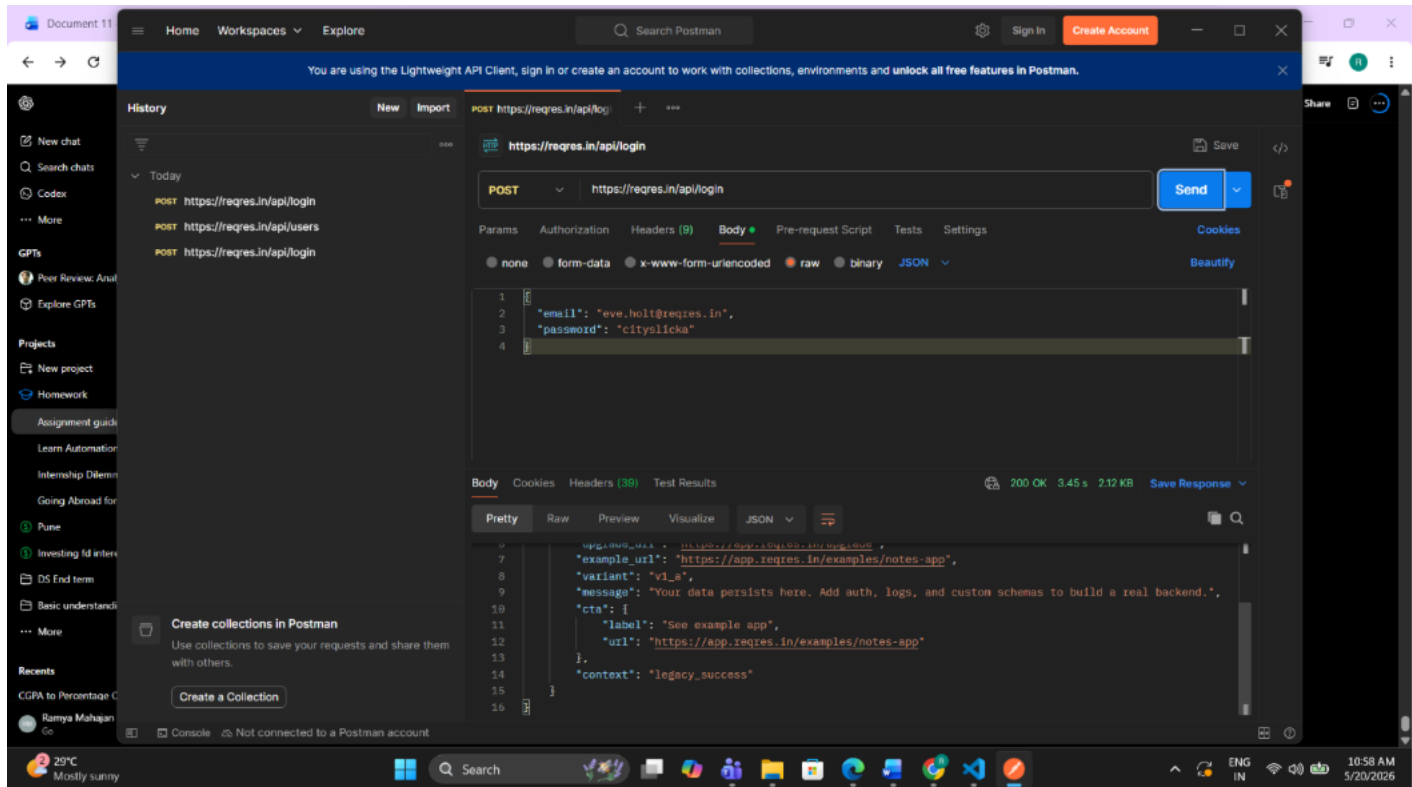
- Method: POST
- Endpoint: `/api/login`

Observation

- API accepted valid credentials successfully.
- Status Code: `200 OK`
- Authentication token generated successfully.
- JSON response structure validated properly.

Screenshot of successful login response showing:

- URL
- JSON body
- `200 OK`
- token response



2. Wrong Password Test

Purpose

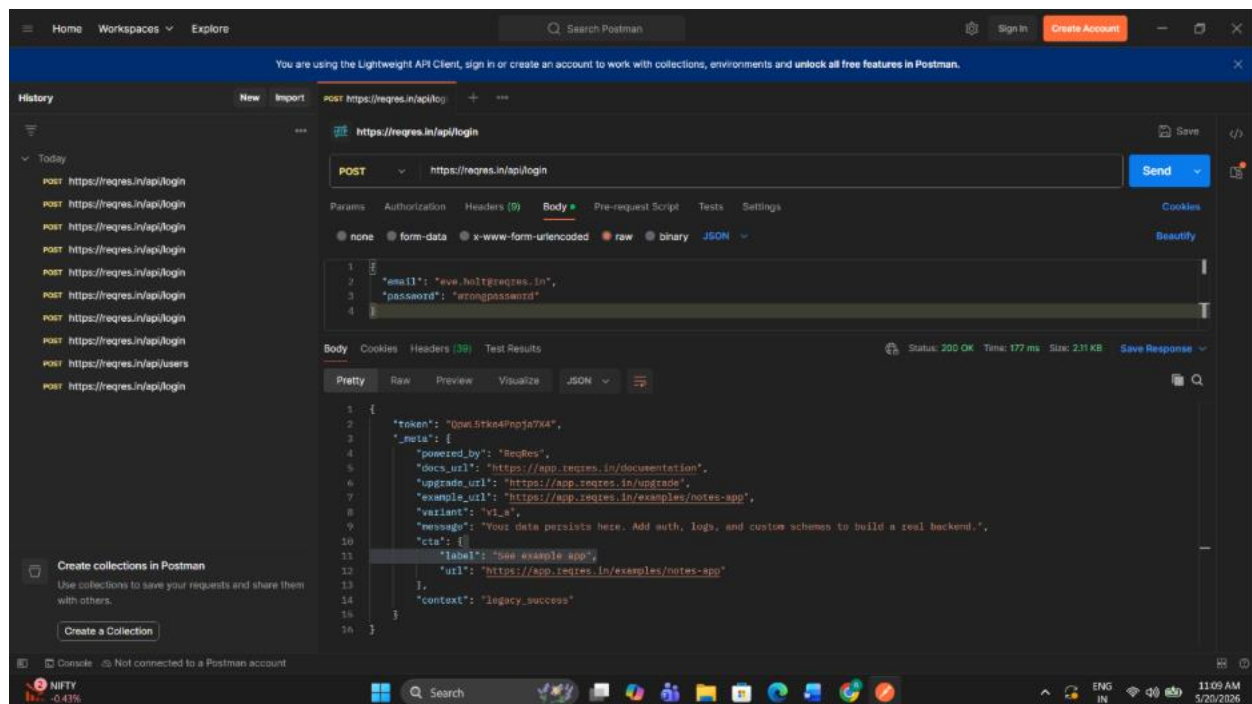
To verify API validation for incorrect password input.

Observation

- API rejected invalid credentials.
- Error handling behavior validated.
- Authentication failure response received.

Screenshot showing:

- wrong password entered
- API error response



3. Empty Request Test

Purpose

To verify API behavior when request body is empty.

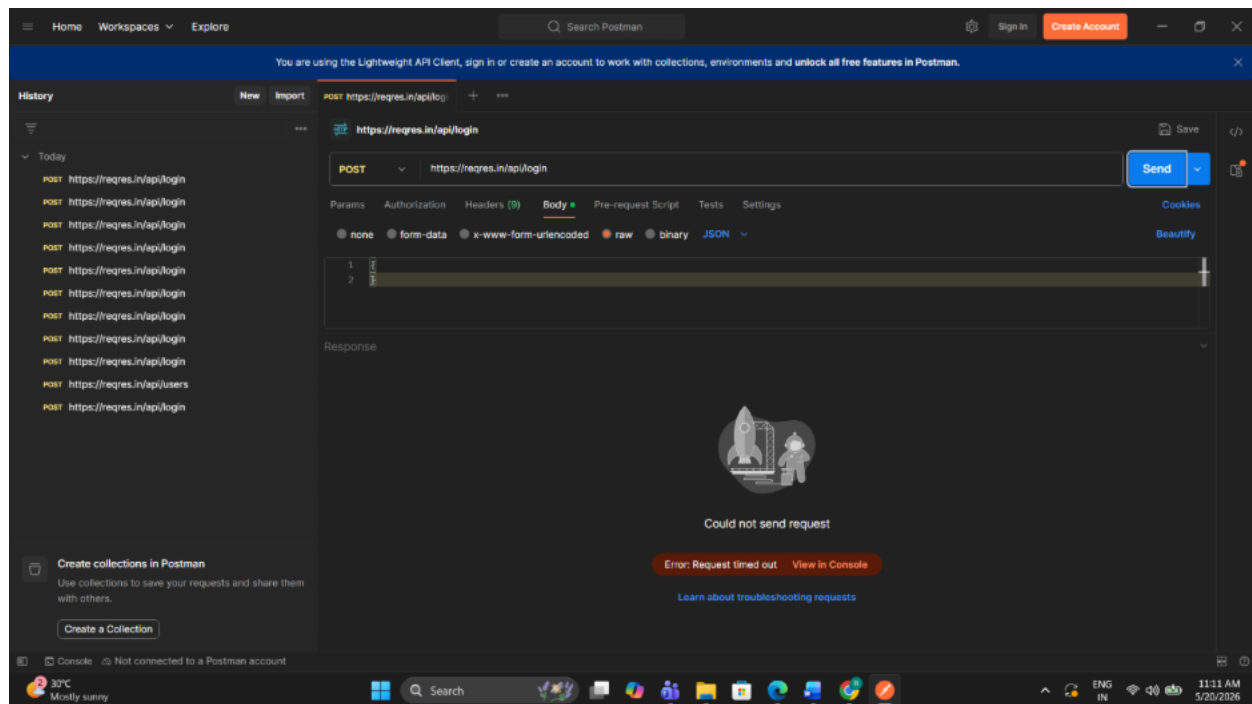
Observation

- API validation failed for empty request body.
- Request rejection behavior observed.

Screenshot showing:

```
{}
```

and response/error message.



4. Invalid Data Test

Purpose

To validate API handling for incorrect datatype values.

Observation

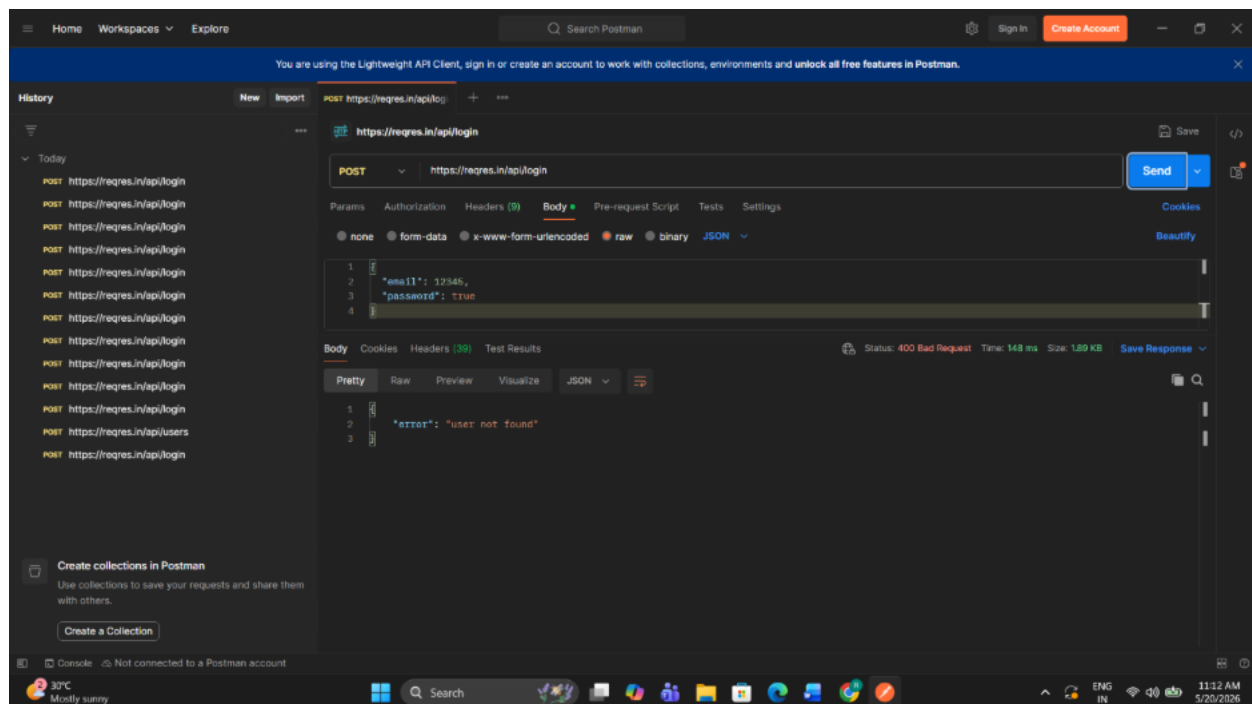
- Invalid JSON values were rejected.
- Status Code: 400 Bad Request
- Validation behavior worked correctly.

Insert Screenshot Here:

Screenshot showing:

```
{  
  "email": 12345,  
  "password": true  
}
```

and 400 Bad Request.



5. Large Input Test

Purpose

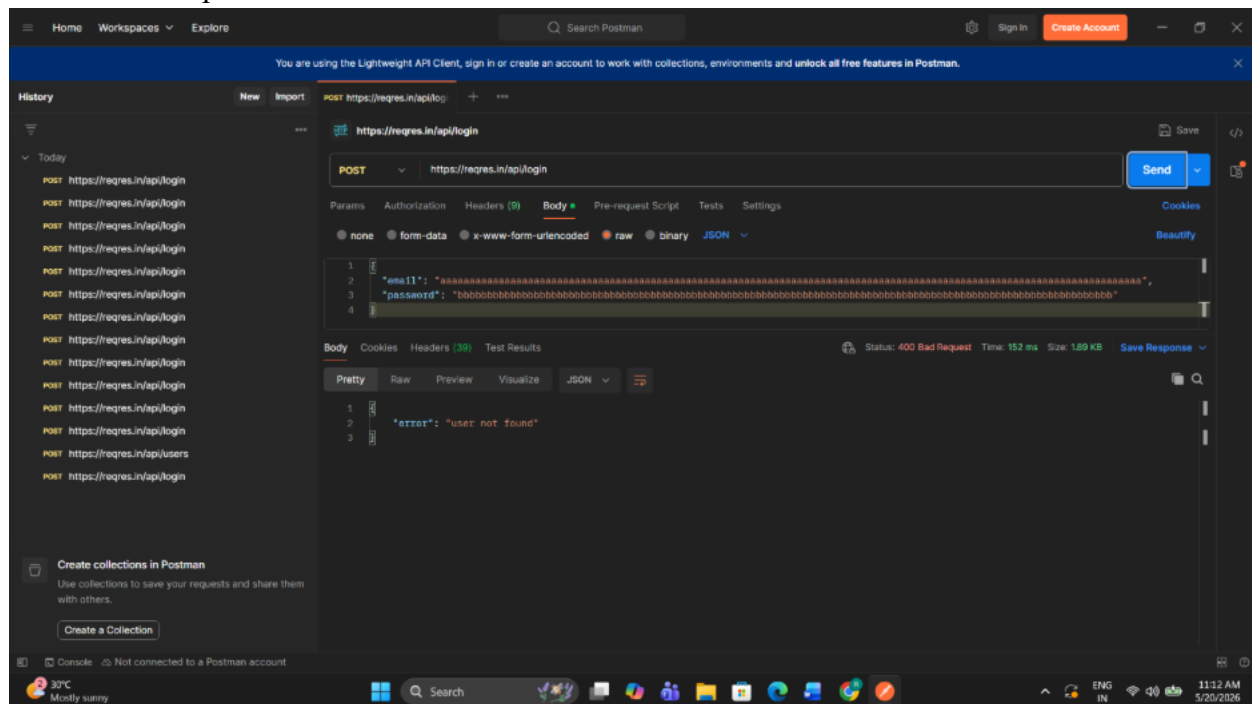
To test API behavior for very large input payloads.

Observation

- Large input values were handled safely.
- Validation error response generated.
- Application remained stable without crash.

Screenshot showing:

- very long email/password values
- API response



Headers & Authentication Validation

Observation

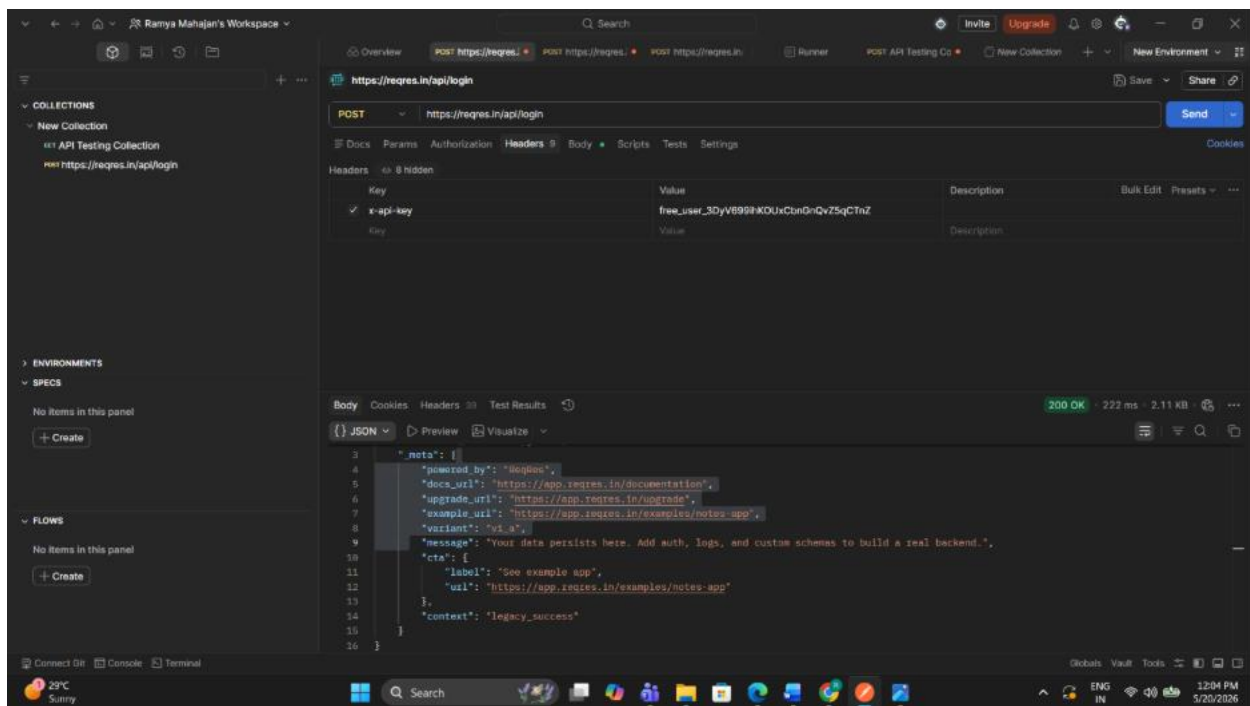
API required authentication using:

x-api-key

Authentication header was added successfully for secured API access.

Screenshot showing:

- Headers tab
- x-api-key



Response Validation Performed

The following API response validations were observed during testing:

- Status codes
- Error messages
- Response time
- JSON structure
- Authentication handling
- Backend validation behavior

Backend Flow Understanding

UI → API → Database → Response

Explanation

1. User sends request from UI.
2. API receives and validates request.
3. Backend/database processes information.
4. Response is returned to the user interface.

Postman Collection Export

API requests were organized inside a Postman collection and exported successfully for future testing and reuse.

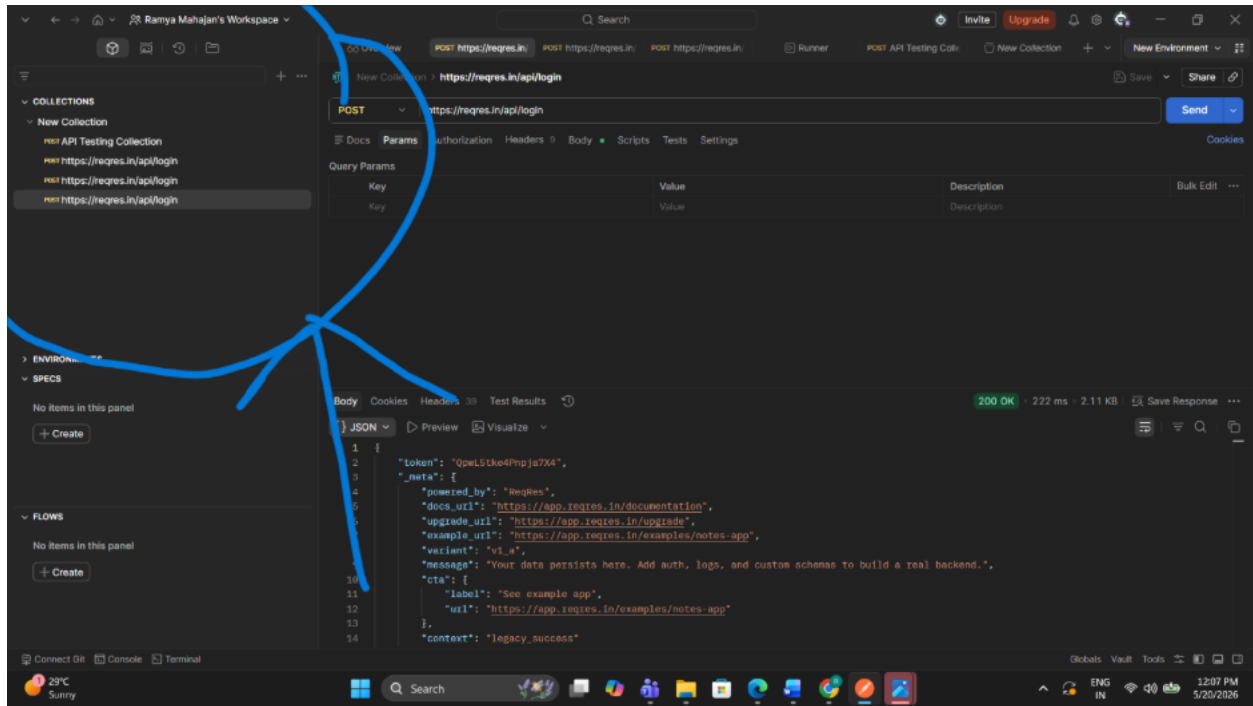
Postman Collection Link:

<https://ramyamahajan05664-2088183.postman.co/workspace/...>

Insert Screenshot Here:

Screenshot of:

- Collection panel
OR
- Export popup
OR
- exported JSON file



API Failure Observations

Observation 1 — Authentication Validation

API required authentication header before processing requests.

Observation 2 — Invalid Credential Handling

Wrong login credentials returned validation failure response.

Observation 3 — Empty Request Validation

API rejected requests with missing input data.

Observation 4 — Invalid Data Handling

Incorrect datatype values returned 400 Bad Request.

Observation 5 — Large Input Handling

Large payload requests were handled safely without application crash.

Conclusion

This API testing activity helped in understanding:

- API request/response workflow
- backend validation process
- authentication handling
- status code verification
- JSON response structure
- error handling mechanisms
- real-world API testing process using Postman